Coupled Cluster Method for Molecular Properties

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Introduction

Motivations

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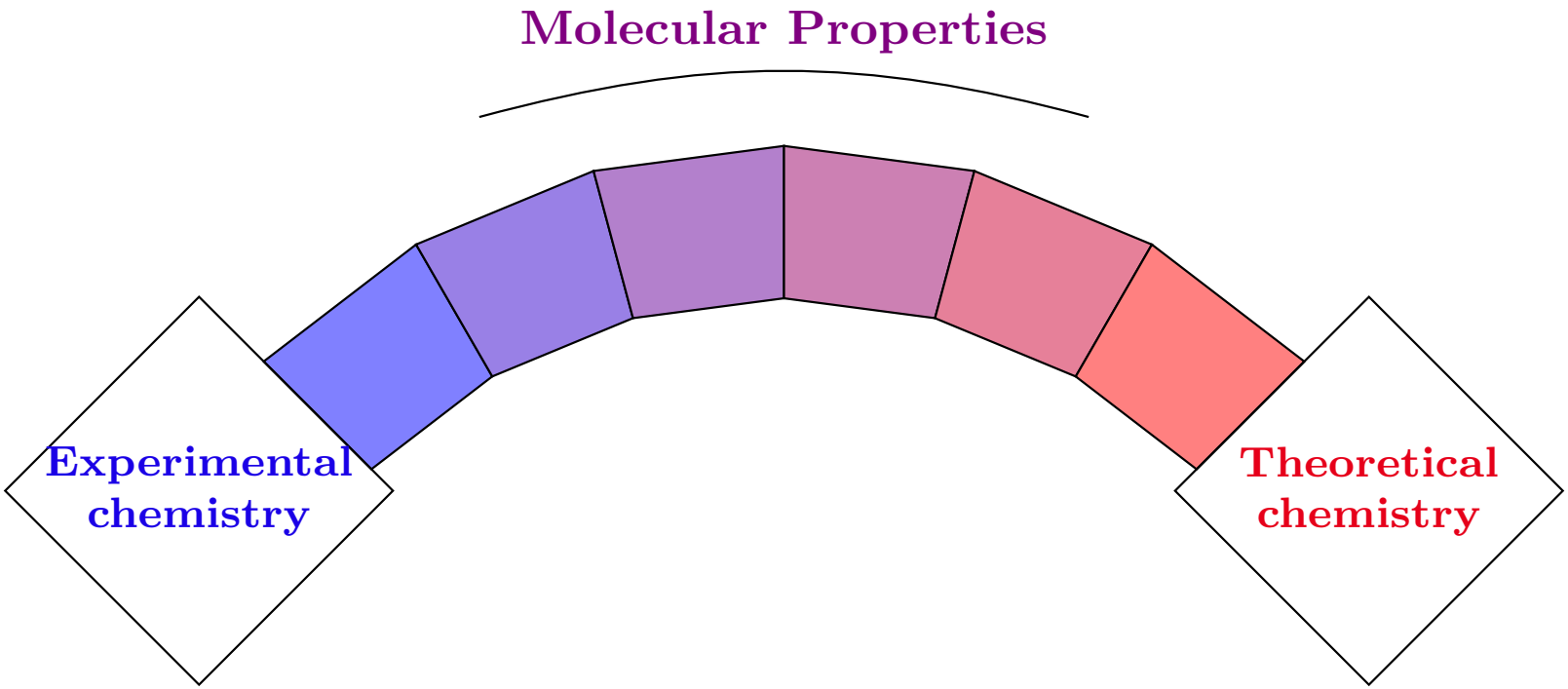
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Molecular Properties

What are they?

$$\hat{H}_0 \tag{1}$$

Unperturbed Hamiltonian

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Time Independent Molecular Properties

$$\underbrace{E^{[1]} = \left.\frac{dE(\varepsilon)}{d\varepsilon}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}}}_{\text{First order}}, \quad \underbrace{E^{[2]} = \left.\frac{d^2E(\varepsilon)}{d\varepsilon^2}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}}}_{\text{Second order}}, \quad \cdots \quad , \underbrace{E^{[n]} = \left.\frac{d^nE(\varepsilon)}{d\varepsilon^n}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}}}_{\text{n-th order}}$$

Molecular properties

A few examples of molecular properties

Electric properties

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Mixed properties

$$\frac{d^2E}{dR_i d\mathcal{E}_\alpha}, \quad \frac{d^2E}{dm_{K_i} dB_\alpha}, \quad \dots$$

Molecular Properties

Feynman, R. P., 1939, Phys. Rev., 56, 340
Hellmann, H., 1937, Einführung in die Quantenchemie (Leipzig: Deuticke)

- The perturbed energy is

$$E(\varepsilon) = \langle \Psi | \hat{H}(\varepsilon) | \Psi \rangle \tag{3}$$

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- The **exact wave function** is needed.



Figure 1: KC Green's web comic strip "On Fire"

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Approximate wave functions

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- First order properties:

Response term

$$\left.\frac{dE(\boldsymbol{\lambda}, \boldsymbol{\varepsilon})}{d\varepsilon_A}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}} = \left.\frac{\partial E(\boldsymbol{\lambda}, \boldsymbol{\varepsilon})}{\partial \varepsilon_A}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}} + \left.\frac{\partial E(\boldsymbol{\lambda}, \boldsymbol{\varepsilon})}{\partial \lambda_\mu} \frac{d\lambda_\mu}{d\varepsilon_A}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}} \tag{6}$$

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- Second order properties:

$$\left.\frac{d^2 E(\boldsymbol{\lambda}, \boldsymbol{\varepsilon})}{d\varepsilon_A d\varepsilon_B}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}} = \left[\frac{\partial^2 E(\boldsymbol{\lambda}, \boldsymbol{\varepsilon})}{\partial \varepsilon_A \partial \varepsilon_B} + \frac{\partial^2 E(\boldsymbol{\lambda}, \boldsymbol{\varepsilon})}{\partial \lambda_\mu \partial \varepsilon_A} \frac{d\lambda_\mu}{d\varepsilon_B}\right]_{\boldsymbol{\varepsilon}=\mathbf{0}} \tag{7}$$

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  - We need the first order of the **response of the wave function**.

$$\left.\frac{d}{d\varepsilon_B}\left(\frac{\partial E(\boldsymbol{\lambda}, \boldsymbol{\varepsilon})}{\partial \lambda_\mu}\right)\right|_{\boldsymbol{\varepsilon}=\mathbf{0}} = 0 \Rightarrow \frac{d\lambda_\mu}{d\varepsilon_B} = \left[-\frac{\partial^2 E(\boldsymbol{\lambda}, \boldsymbol{\varepsilon})}{\partial \varepsilon_B \partial \lambda_\mu}\left(\frac{\partial^2 E(\boldsymbol{\lambda}, \boldsymbol{\varepsilon})}{\partial \lambda_\mu^2}\right)^{-1}\right]_{\boldsymbol{\varepsilon}=\mathbf{0}} \tag{8}$$

Coupled Cluster Theory

A non complete introduction

- In coupled-cluster theory we use a non-unitary exponential parametrization:

Exponential Cluster operator

$$|\Psi_{\mathrm{CC}}\rangle = e^{\hat{T}} |\Phi_0\rangle \tag{9}$$

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- By inserting the exponential ansatz into the Schrödinger equation, we obtain the energy:

$$\hat{H} \, |\Psi_{\mathrm{CC}}\rangle = E_{\mathrm{CC}} \, |\Psi_{\mathrm{CC}}\rangle \Rightarrow E_{\mathrm{CC}} = \frac{\langle \Psi_{\mathrm{CC}} | \hat{H} | \Psi_{\mathrm{CC}} \rangle}{\langle \Psi_{\mathrm{CC}} | \Psi_{\mathrm{CC}} \rangle} \tag{10}$$

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- In CC community we use the **similarity-transformed Hamiltonian**:

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- Projecting

$$\langle \Phi_0 | \quad : \quad \langle \Phi_0 | \hat{\overline{H}} | \Phi_0 \rangle = E_{\mathrm{CC}} \leftarrow \text{Energy (not an upper bound)} \tag{11}$$

$$\langle \Phi_\mu | \quad : \quad \langle \Phi_\mu | \hat{\overline{H}} | \Phi_0 \rangle = 0 \leftarrow \text{Amplitudes} \tag{12}$$

Molecular Properties through Coupled Cluster Theory

Energy derivatives for non-variational wave functions

- We have

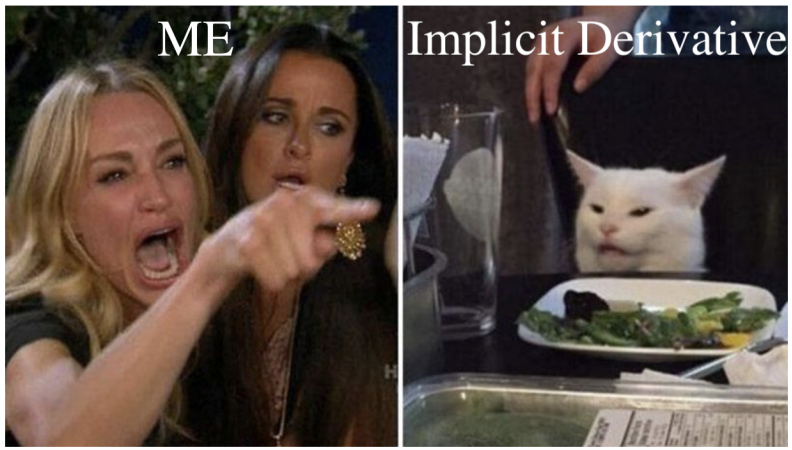
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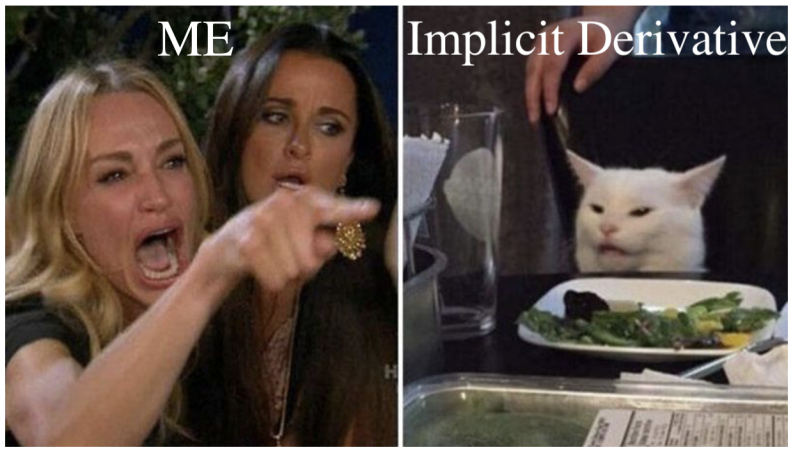


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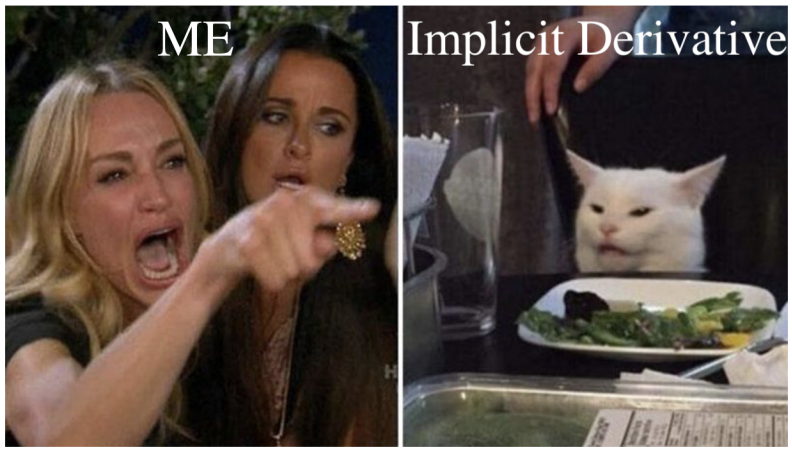
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- In principle we should compute $\frac{dt_\nu}{d\varepsilon_A}$ for *each* perturbation strength.
  - So much effort to just have the gradient... **Can we do better?**

Molecular Properties through Coupled Cluster Theory

T.Helgaker and P.Jørgensen. *Theor. Chim. Acta*, 75:111, 1989

- We introduce the CC Lagrangian:

$$L_{\mathrm{CC}}(\boldsymbol{t}, \bar{\boldsymbol{t}}, \boldsymbol{\varepsilon}) = E_{\mathrm{CC}}(\boldsymbol{t}, \boldsymbol{\varepsilon}) + \sum_{\mu} \bar{t}_{\mu} \langle \Phi_{\mu} | \hat{\bar{H}} | \Phi_0 \rangle \tag{14}$$

Multipliers

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$$\left.\frac{\partial L_{\mathrm{CC}}(\boldsymbol{t},\bar{\boldsymbol{t}},\boldsymbol{\varepsilon})}{\partial \bar{t}_\nu}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}} = 0 = \langle \Phi_\nu | \hat{\overline{H}} | \Phi_0 \rangle \leftarrow \text{Amplitudes} \tag{16}$$

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$$\left.\frac{dL_{\mathrm{CC}}(\boldsymbol{t},\bar{\boldsymbol{t}},\boldsymbol{\varepsilon})}{d\varepsilon_A}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}} = \left[\frac{\partial L_{\mathrm{CC}}(\boldsymbol{t},\bar{\boldsymbol{t}},\boldsymbol{\varepsilon})}{\partial \varepsilon_A} + \frac{\partial L_{\mathrm{CC}}(\boldsymbol{t},\bar{\boldsymbol{t}},\boldsymbol{\varepsilon})}{\partial t_\nu}\frac{dt_\nu}{d\varepsilon_A} + \frac{\partial L_{\mathrm{CC}}(\boldsymbol{t},\bar{\boldsymbol{t}},\boldsymbol{\varepsilon})}{\partial \bar{t}_\nu}\frac{d\bar{t}_\nu}{d\varepsilon_A}\right]_{\boldsymbol{\varepsilon}=\mathbf{0}} \tag{15}$$

  - We impose the stationarity of the Lagrangian with respect to t_ν and $\bar{t}_\nu$:

$$\left.\frac{\partial L_{\mathrm{CC}}(\boldsymbol{t},\bar{\boldsymbol{t}},\boldsymbol{\varepsilon})}{\partial \bar{t}_\nu}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}} = 0 = \langle \Phi_\nu | \hat{\overline{H}} | \Phi_0 \rangle \leftarrow \text{Amplitudes} \tag{16}$$

$$\left.\frac{\partial L_{\mathrm{CC}}(\boldsymbol{t},\bar{\boldsymbol{t}},\boldsymbol{\varepsilon})}{\partial t_\nu}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}} = 0 = \langle \Phi_0 | (1+\hat{\Lambda})[\hat{\overline{H}}, \hat{\tau}_\nu] | \Phi_0 \rangle \leftarrow \text{Multipliers} \tag{17}$$

  - We conclude that:

$$\left.\frac{dL_{\mathrm{CC}}(\boldsymbol{t},\bar{\boldsymbol{t}},\boldsymbol{\varepsilon})}{d\varepsilon_A}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}} = \left.\frac{\partial L_{\mathrm{CC}}(\boldsymbol{t},\bar{\boldsymbol{t}},\boldsymbol{\varepsilon})}{\partial \varepsilon_A}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}} = \left[\langle \Phi_0 | (1+\hat{\Lambda}) \frac{\partial \hat{\overline{H}}}{\partial \varepsilon_A} | \Phi_0 \rangle\right]_{\boldsymbol{\varepsilon}=\mathbf{0}} \tag{18}$$

tenpi

Elegant development of CC methods

- General-order CC generator based on diagrams.
- Automatic optimization of intermediates available.
- Export compilable code (Fortran, ...) ⇒ Integration with **DIRAC** (`http://diracprogram.org`).

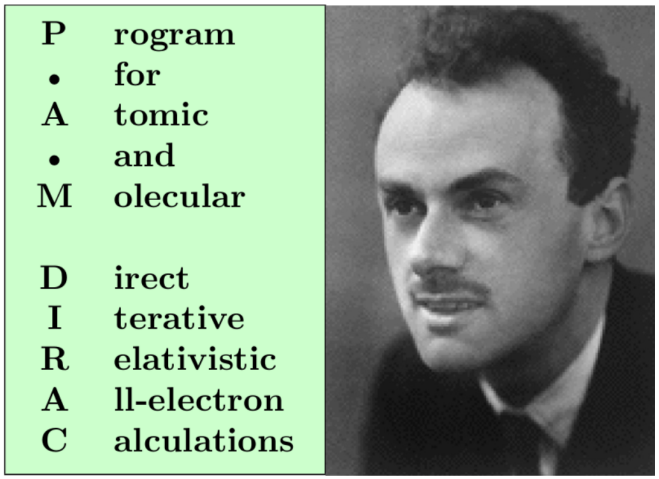
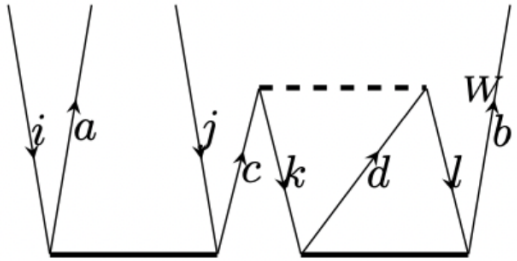
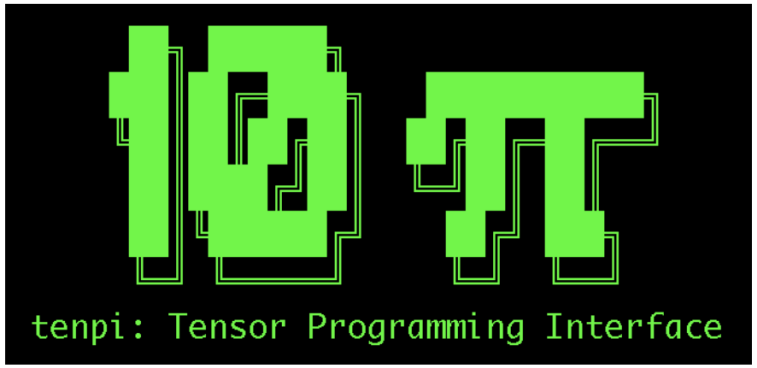


Figure 2: L. Visscher, H. J. Aa. Jensen, R. Bast, A. S. P. Gomes and T. Saue *et al.*



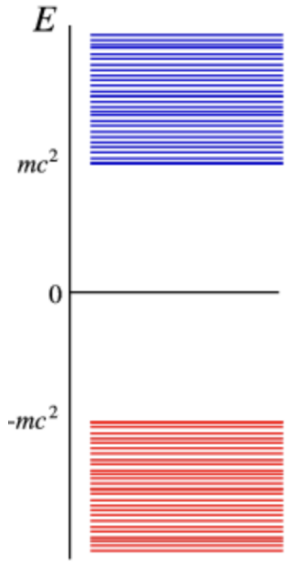
$$\frac{1}{2}\, P(a/b)\, V_{cd}^{kl}\, t_{ij}^{ac}\, t_{kl}^{db}$$

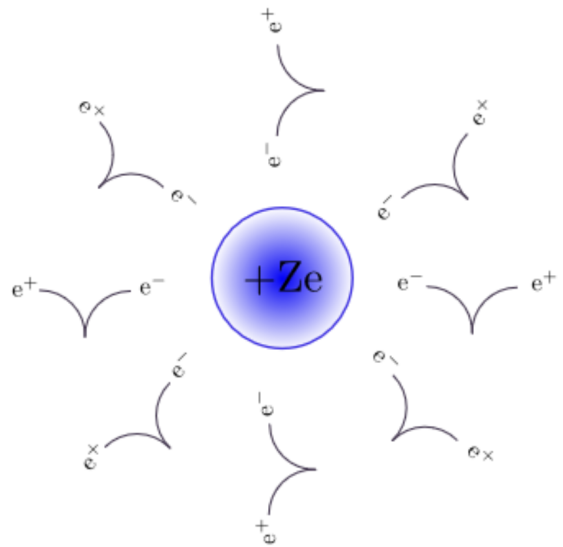


Main author: Jan Brandejs

Conclusions and Perspectives

1. CC development
   - Perturbative approaches i.e CCSD(T).
   - Lambda (multipliers) equations.
2. Molecular Properties
   - Magnetic properties require the use of **negative energy solutions** of the Dirac equation: **challenging**.
   - Core molecular properties: where **QED-effects** may play an important role.





HAMP-vQED

Highly Accurate Molecular Properties using Variational Quantum Electrodynamics

HAMP-vQED team








From the left: Jan Brandejs, Gabriele Fabbro, Trond Saue, Ryan Benazzouk and Dávid Ferenc.